

Jindong Cao

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Education

Carnegie Mellon University, College of Engineering
M.S. in Electrical and Computer Engineering, GPA: 3.75/4.0

Pittsburgh, USA
Aug 2023 – Dec 2024

Sun Yat-sen University, School of Computer Science and Engineering
B.S. in Computer Science and Technology, GPA: 3.9/4.0

Guangzhou, China
Aug 2019 – Jul 2023

Experience

TikTok
Machine Learning Engineer

San Jose, USA
Jan 2025 – Present

- Owned end-to-end push CTR (Click-Through-Rate) modeling pipeline, including feature engineering (User-side, Content-side features), label construction, model design (TokenMixer, MMCN, Multi-Task Learning), datastream design and online serving; launched model and scheduling updates that cumulatively increased push clicks by **3%** and drove **0.2% DAU lift**.
- Designed and launched TikTok E-Commerce push recommendation models, collaborating with architecture teams to build the E-Commerce push retrieval and ranking pipeline; led feature engineering, CTR model design, and template-selection model design, driving a **1.2%** lift in TikTok Shop GMV.
- Redesigned the core push scheduling architecture from FIFO-based delivery to dynamic candidate retrieval and global value-based ranking; implemented a unified C++ framework to compare candidates across scenarios at request time, improving pushed-video finish rate by **6%**.
- Built a VLM-based user-interest discovery pipeline that fuses cross-modal and cross-scenario input signals to construct semantic IDs and generate interest candidates; offline evaluation demonstrated improved interest coverage and candidate diversity beyond historical engagement signals.

Amazon
Software Development Engineer Intern

San Diego, USA
May 2024 – Aug 2024

- Built internal full-stack tools with React, Java, and TypeScript, improving service accessibility and operational efficiency.

ZMO.AI
Machine Learning Engineer Intern

Shenzhen, China
Mar 2023 – May 2023

- Developed an AI image editing pipeline in Python integrating ChatGPT, SAM, GroundingDINO, and Stable Diffusion.
- Applied pruning to BLIP-2 and CLIP, reducing memory usage by 30% and runtime by 40%.

Huawei Technologies
Software Development Engineer Intern

Dongguan, China
Jul 2022 – Sep 2022

- Refactored a data center query system in C++, reducing average query latency by 70%.

Research

Weakly supervised prompt learning for low-resource medical image classification, Pattern Recognition
Co-author Jan 2024

- Leveraged transferable representations from large-scale pre-trained vision-language models to address low-resource medical image classification for clinical decision support.
- Improved zero-shot and few-shot classification performance via additional prompt-embedding structures.

Skills

Python, C++, SQL, PyTorch, TensorFlow, Deep Learning, Machine Learning, Recommendation System